

- 3 The ionisation energies, in kJ mol^{-1} , of five elements are given in the table.

	2nd ionisation energy	3rd ionisation energy	4th ionisation energy
F	3370	6040	8410
Ne	3950	6150	9290
Na	4560	6940	9540
Mg	1450	7740	10 500
Al	1820	2740	11 600

Which statement about these ionisation energies is correct?

- A** The 2nd ionisation energy of F is greater than the 3rd ionisation energy of Al because Al^{2+} ions have more outer shell electrons than F^+ ions.
- B** The 3rd ionisation energy of all the elements in the table involves the removal of an electron from the same shell.
- C** The 4th ionisation energy of Na is greater than the 3rd ionisation energy of Ne because the nuclear charge of Na is greater than that of Ne.
- D** The successive ionisation energies of these elements increase as the electrons are being taken from the same shell.